Project APEX Field Workbook

20 labs • 15 debugging investigations • paper-to-code workshops • independent design records

This workbook is intentionally write-heavy. Complete it beside the seven-volume curriculum. A page is not finished until it contains a prediction, runtime evidence, an explanation, a deliberately introduced failure, a repair and an engineering decision.

Evidence Standard

- Prediction written before execution
- Command or code version recorded
- Observed shapes, values or plots captured
- Failure localized to the earliest contract
- Repair protected by a regression test
- Implementation choice justified against an alternative

Lab Worksheet 1: 01 Units And Shapes

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 2: 02 Vectorized Kinematics

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 3: 03 Force Model

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 4: 04 Sampling Aliasing

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 5: 05 Regression Baseline

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 6: 06 Autograd Drag

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 7: 07 Pytorch Module

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 8: 08 Dataset Windows

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 9: 09 Loss Comparison

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 10: 10 Training Loop

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 11: 11 Rnn From Scratch

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 12: 12 Gru Gates

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 13: 13 Linear Ssm

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 14: 14 Selective Ssm

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 15: 15 Autoencoder

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 16: 16 Vae Reparameterization

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 17: 17 Rssm Step

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 18: 18 Cem Planning

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 19: 19 Time Alignment

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Lab Worksheet 20: 20 Pipeline Contract

Before you run

Prediction	Your answer

Runtime trace

Line / operation	Input	Output	Meaning

Break and repair

Injected defect:

Earliest failed invariant:

Regression test:

Decision record

Question	Evidence-based response

Debugging Investigation 1: Unit Mismatch

Speed is converted twice and rollout acceleration is wrong.

Do not repair yet

1. What should be invariant?
2. What is the earliest component that can violate it?
3. What is the smallest input that reproduces the problem?
4. Which probe distinguishes your top two hypotheses?
5. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 2: Axis Swap

A [batch,time,feature] tensor is passed as [time,batch,feature].

Do not repair yet

6. What should be invariant?

7. What is the earliest component that can violate it?

8. What is the smallest input that reproduces the problem?

9. Which probe distinguishes your top two hypotheses?

10. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 3: Target Leakage

Future speed accidentally appears in model inputs.

Do not repair yet

11. What should be invariant?

12. What is the earliest component that can violate it?

13. What is the smallest input that reproduces the problem?

14. Which probe distinguishes your top two hypotheses?

15. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 4: Random Window Split

Overlapping windows from one session enter every split.

Do not repair yet

16. What should be invariant?

17. What is the earliest component that can violate it?

18. What is the smallest input that reproduces the problem?

19. Which probe distinguishes your top two hypotheses?

20. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 5: Stale Asof Join

Weather is forward-filled across a long outage.

Do not repair yet

21. What should be invariant?

22. What is the earliest component that can violate it?

23. What is the smallest input that reproduces the problem?

24. Which probe distinguishes your top two hypotheses?

25. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 6: Normalizer Leak

Standardization is fitted on all data.

Do not repair yet

26. What should be invariant?

27. What is the earliest component that can violate it?

28. What is the smallest input that reproduces the problem?

29. Which probe distinguishes your top two hypotheses?

30. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 7: Hidden State Reuse

Hidden state leaks between unrelated sessions.

Do not repair yet

31. What should be invariant?

32. What is the earliest component that can violate it?

33. What is the smallest input that reproduces the problem?

34. Which probe distinguishes your top two hypotheses?

35. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 8: Missing Eval Mode

Dropout remains active during evaluation.

Do not repair yet

36. What should be invariant?

37. What is the earliest component that can violate it?

38. What is the smallest input that reproduces the problem?

39. Which probe distinguishes your top two hypotheses?

40. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 9: Gradient Accumulation

Gradients are never cleared.

Do not repair yet

41. What should be invariant?

42. What is the earliest component that can violate it?

43. What is the smallest input that reproduces the problem?

44. Which probe distinguishes your top two hypotheses?

45. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 10: Exploding Ssm

Unconstrained recurrent decay exceeds one.

Do not repair yet

46. What should be invariant?

47. What is the earliest component that can violate it?

48. What is the smallest input that reproduces the problem?

49. Which probe distinguishes your top two hypotheses?

50. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 11: Kl Collapse

RSSM posterior ignores stochastic state.

Do not repair yet

51. What should be invariant?

52. What is the earliest component that can violate it?

53. What is the smallest input that reproduces the problem?

54. Which probe distinguishes your top two hypotheses?

55. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 12: Planner Exploitation

CEM finds actions outside training support.

Do not repair yet

56. What should be invariant?

57. What is the earliest component that can violate it?

58. What is the smallest input that reproduces the problem?

59. Which probe distinguishes your top two hypotheses?

60. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 13: Artifact Overwrite

Two runs write the same model path.

Do not repair yet

61. What should be invariant?

62. What is the earliest component that can violate it?

63. What is the smallest input that reproduces the problem?

64. Which probe distinguishes your top two hypotheses?

65. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 14: Airflow Payload

Large arrays are passed through XCom.

Do not repair yet

66. What should be invariant?

67. What is the earliest component that can violate it?

68. What is the smallest input that reproduces the problem?

69. Which probe distinguishes your top two hypotheses?

70. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Debugging Investigation 15: Ui False Precision

The UI reports more precision than the model supports.

Do not repair yet

71. What should be invariant?

72. What is the earliest component that can violate it?

73. What is the smallest input that reproduces the problem?

74. Which probe distinguishes your top two hypotheses?

75. What downstream symptom could distract you?

Repair record

Item	Your evidence

Regression test specification

Given:

When:

Then:

Paper-to-Code Workshop 1: DreamerV3

Reference: arXiv:2301.04104

Focus: World model, reward/continuation heads, imagined actor-critic learning and robustness transformations.

One-page paper decomposition

Question	Your implementation interpretation
What problem is being solved?	
What is the input and target?	
What information is hidden or latent?	
What loss terms exist?	
What prevents collapse or exploitation?	
What is the inference-time path?	
Which baseline is load-bearing?	
Which claim can APEX test cheaply?	

Code mapping

Write the modules, tensors and tests required for the smallest faithful reproduction.

Paper component	APEX module / tensor / test

Paper-to-Code Workshop 2: Mamba

Reference: arXiv:2312.00752

Focus: Input-dependent selective state-space dynamics and hardware-aware sequence computation.

One-page paper decomposition

Question	Your implementation interpretation
What problem is being solved?	
What is the input and target?	
What information is hidden or latent?	
What loss terms exist?	
What prevents collapse or exploitation?	
What is the inference-time path?	
Which baseline is load-bearing?	
Which claim can APEX test cheaply?	

Code mapping

Write the modules, tensors and tests required for the smallest faithful reproduction.

Paper component	**APEX module / tensor / test**

Paper-to-Code Workshop 3: I-JEPA

Reference: arXiv:2301.08243

Focus: Context-to-target representation prediction without pixel reconstruction.

One-page paper decomposition

Question	Your implementation interpretation
What problem is being solved?	
What is the input and target?	
What information is hidden or latent?	
What loss terms exist?	
What prevents collapse or exploitation?	
What is the inference-time path?	
Which baseline is load-bearing?	
Which claim can APEX test cheaply?	

Code mapping

Write the modules, tensors and tests required for the smallest faithful reproduction.

Paper component	**APEX module / tensor / test**

Paper-to-Code Workshop 4: LeWorldModel

Reference: arXiv:2603.19312

Focus: End-to-end reconstruction-free world-model learning with embedding prediction and latent regularization.

One-page paper decomposition

Question	Your implementation interpretation
What problem is being solved?	
What is the input and target?	
What information is hidden or latent?	
What loss terms exist?	
What prevents collapse or exploitation?	
What is the inference-time path?	
Which baseline is load-bearing?	
Which claim can APEX test cheaply?	

Code mapping

Write the modules, tensors and tests required for the smallest faithful reproduction.

Paper component	**APEX module / tensor / test**

Paper-to-Code Workshop 5: F1 25 UDP Specification

Reference: EA Data Output from F1 25 Game v3

Focus: Packed little-endian packet contracts, frame synchronization, rates, privacy restrictions and replay requirements.

One-page paper decomposition

Question	Your implementation interpretation
What problem is being solved?	
What is the input and target?	
What information is hidden or latent?	
What loss terms exist?	
What prevents collapse or exploitation?	
What is the inference-time path?	
Which baseline is load-bearing?	
Which claim can APEX test cheaply?	

Code mapping

Write the modules, tensors and tests required for the smallest faithful reproduction.

Paper component	**APEX module / tensor / test**

Original World-Model Design Studio

Do not begin with an architecture name. Begin with a measured failure in APEX.

1. Failure definition

76. Which held-out situation fails?

77. What metric and horizon expose it?

78. What simpler explanation has been ruled out?

2. State design

79. Which variables belong in state, action, context and observation?

80. What must be latent?

81. What must remain physically interpretable?

3. Memory design

82. What duration matters?

83. Is access content-based, recurrent or hierarchical?

84. What is the compute budget?

4. Objective design

85. What should be predicted?

86. Which loss terms are necessary?

87. How will collapse or exploitation be detected?

5. Evaluation design

88. What baseline must be beaten?

89. Which ablation isolates the contribution?

90. What physical guardrails block promotion?

Final Independent Build Checklist

☐ Canonical contract created from memory

☐ Group split proven leakage-free

☐ Baseline reproduced

☐ GRU rollout rebuilt

☐ Selective SSM challenger implemented

☐ RSSM prior/posterior paths explained

☐ CEM planner constrained

☐ Ablations recorded

☐ Pipeline stages idempotent

☐ API contract versioned

☐ UI communicates uncertainty

☐ F1 25 replay adapter designed

☐ One original architecture proposed and falsifiable